

State-space model and the Kalman filter

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1 Recap

- We now know some of the fundamentals of time series analysis
- We know well the difference between stationary and non-stationary variables and time series processes, such as:
 - AR, MA, RW
- We also know why we sometimes want to focus on trend and cycle decompositions and how to compute them using different methods
- We also now know how to formulate, estimate, and predict the future with a popular type of multivariate time series model
 - VAR

2 Why state-space and Kalman filter?

- Highly useful tool for nearly all inference problems in macroeconomics and finance:
 - Can be applied to univariate and multivariate problems
 - Can be applied to stationary and non-stationary processes
 - Can infer latent states and missing values
- Originally developed to track rockets lol

3 State-space representation

Observation equation:

$$y_t = H\beta_t + Az_t + e_t$$

Transition equation:

$$\beta_t = \tilde{\mu} + F\beta_{t-1} + v_t$$

with

$$\begin{bmatrix} e_t \\ v_t \end{bmatrix} \sim \text{i.i.d.} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} R & 0 \\ 0 & Q \end{bmatrix} \right)$$

Definitions:

- y_t : $(n \times 1)$ vector of observable variables
- β_t : $(k \times 1)$ vector of (unobserved) state variables
- z_t : $(r \times 1)$ vector of exogenous observed variables
- e_t, v_t : $(n \times 1)$ and $(k \times 1)$ vectors of measurement and transition equation errors
- $H, A, F, \tilde{\mu}$: $(n \times k)$, $(n \times r)$, $(k \times k)$, and $(k \times 1)$ parameter matrices
- R, Q : $(n \times n)$ and $(k \times k)$ covariance matrices

The observation equation is just n linear equations

- Often, but not always, R is assumed to be diagonal, i.e., n independent linear static equations

The transition equation follows a VAR structure. Note: R and Q are assumed to be independent (can be relaxed, but not now).

3.1 VAR models in state-space form

- Nearly all linear models can be written in state-space form (VAR, ARMA, UC etc.)
- Consider the VAR model:

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.5 & 0 \\ 1 & 0.2 \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix}$$

with

$$\begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix} \sim \text{i.i.d.} N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix} \right)$$

Written in a state-space representation we get the measurement equation:

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_{1,t} \\ \beta_{2,t} \end{bmatrix} \quad (1)$$

and the associated transition equation:

$$\begin{bmatrix} \beta_{1,t} \\ \beta_{2,t} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.5 & 0 \\ 1 & 0.2 \end{bmatrix} \begin{bmatrix} \beta_{1,t-1} \\ \beta_{2,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix}$$

Accordingly, $n = 2$, $k = 2$, $r = 0$ and:

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \tilde{\mu} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad F = \begin{bmatrix} 0.5 & 0 \\ 1 & 0.2 \end{bmatrix} \quad Q = \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix}$$

$z_t = 0$ and $e_t = 0 \forall t$, i.e., no exogenous variables and no measurement errors

- But, in state-space form, more than one way of rewriting a model
- In the case just described we could for example have set $r = 1$, and let $\tilde{\mu} = [0 \ 0]'$, yielding the state-space representation:

$$\begin{aligned} \begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_{1,t} \\ \beta_{2,t} \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ \begin{bmatrix} \beta_{1,t} \\ \beta_{2,t} \end{bmatrix} &= \begin{bmatrix} 0.5 & 0 \\ 1 & 0.2 \end{bmatrix} \begin{bmatrix} \beta_{1,t-1} \\ \beta_{2,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix} \end{aligned} \quad (2)$$

In (1) $\beta_{i,t} = y_{i,t}$, while in (2) $\beta_{i,t} = y_{i,t} - \tilde{\mu}_i = y_{i,t} - 1$ for $i = 1, 2$

3.2 ARMA models in state-space form

Consider the ARMA(1, 1) model:

$$y_t = \phi_1 y_{t-1} + \epsilon_t + \theta \epsilon_{t-1} \Rightarrow (1 - \theta L)y_t = (1 + \theta L)\epsilon_t$$

Put in state-space form

$$y_t = \begin{bmatrix} 1 & \theta \end{bmatrix} \begin{bmatrix} \beta_{1,t} \\ \beta_{2,t} \end{bmatrix} \text{ and } \begin{bmatrix} \beta_{1,t} \\ \beta_{2,t} \end{bmatrix} = \begin{bmatrix} \phi & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \beta_{1,t-1} \\ \beta_{2,t-1} \end{bmatrix} + \begin{bmatrix} \epsilon_t \\ 0 \end{bmatrix}$$

with first line from transition equation and observation equation

$$(1 - \phi L)\beta_{1,t} = \epsilon_t \text{ and } y_t = (1 + \theta L)\beta_{1,t}$$

since $\beta_{2,t} = \beta_{1,t-1}$. Thus

$$y_t = (1 + \theta L)(1 - \phi L)^{-1}\epsilon_t \Rightarrow (1 - \phi L)y_t = (1 + \theta L)\epsilon_t$$

3.3 Trends and cycle methods recap

- In earlier lectures we talked about different methods for directly estimating the trend and cycle from non-stationary data
- A class of models that are often used for this is called Unobserved Components (UC) models
- As the name suggests, these models contain unobserved components
- The state-space formulation and the Kalman Filter are often used for estimation

3.4 Unobserved components models

Consider the following simple model

$$y_t = g_t + c_t$$

where g_t is the trend and c_t is the cycle

Assume also that the stochastic trend is driven by a random walk with drift:

$$g_t = \tilde{\mu}_1 + g_{t-1} + \epsilon_{1,t}$$

and the cyclical component follows a stationary AR(2) model:

$$c_t = \phi_1 c_{t-1} + \phi_2 c_{t-2} + \epsilon_{2,t}$$

where

$$\begin{bmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \end{bmatrix} \sim \text{i.i.d.} N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \right) \quad (3)$$

Employing the state-space representation we get the following measurement and transition equations:

$$y_t = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \beta_{1,t} \\ \beta_{2,t} \\ \beta_{2,t-1} \end{bmatrix}$$

so that, $\beta_{1,t} = g_t$ and $\beta_{2,t} = c_t$

$$\begin{bmatrix} \beta_{1,t} \\ \beta_{2,t} \\ \beta_{2,t-1} \end{bmatrix} = \begin{bmatrix} \tilde{\mu} \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & \phi_1 & \phi_2 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \beta_{1,t-1} \\ \beta_{2,t-1} \\ \beta_{2,t-2} \end{bmatrix} + \begin{bmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \\ 0 \end{bmatrix}$$

Here, $n = 1$, $k = 3$, $r = 0$, and:

$$H = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \quad \tilde{\mu} \begin{bmatrix} \tilde{\mu}_1 \\ 0 \end{bmatrix} \quad F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \phi_1 & \phi_2 \\ 0 & 1 & 0 \end{bmatrix} \quad A = 0$$

and $e_t = 0 \forall t$. Q^* is given by the covariance matrix in (3), implying that:

$$G = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad Q = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3.5 State-space form: don't forget

- The state variables do not have to be unobserved in the state-space representation
- There are usually many ways of writing a model into state-space form
- Why do we do it?
 - Facilitates employing the Kalman Filter
 - Facilitates MLE
 - Facilitates (conditional) forecasting and inferring missing values
- BUT: subtle issue regarding for identification (not finna look at it :skull:)

4 The Kalman filter

- Once a model is put into state-space form we can utilize the Kalman filter machinery
- The Kalman filter is:
 - A linear filter
 - A recursive procedure for computing the optimal estimate of the unobserved state vector
 - Provides the minimum squared error estimate of the unobserved state irrespective of the assumptions on the error terms (R and Q) – if Gaussian, the filter provides the exact conditional probability
- The Kalman filter can also be used for MLE
- Consists of two (three) steps: prediction and updating done recursively through time. Once at the end of a sample (step three) smoothing can be applied

4.1 Some useful definitions

- $\beta_{t|t-1}$: expectation of β_t conditional on information up to $t-1$, i.e., $\mathbb{E}[\beta_t | I_{t-1}]$
- $P_{t|t-1}$: covariance matrix of β_t conditional on information up to $t-1$, i.e., $\mathbb{E}[(\beta_t - \beta_{t|t-1})(\beta_t - \beta_{t|t-1})']$
- $\beta_{t|t}$: expectation of β_t conditional on information up to t
- $P_{t|t}$: covariance matrix of β_t conditional on information up to t , i.e., $\mathbb{E}[(\beta_t - \beta_{t|t})(\beta_t - \beta_{t|t})']$
- $y_{t|t-1}$: forecast of y_t given information up to time $t-1$
- $\eta_{t|t-1}$: forecast error of y_t , i.e., $y_t - y_{t|t-1}$
- $f_{t|t-1}$: conditional covariance of the forecast error, $\eta_{t|t-1}$, i.e., $\mathbb{E}[(y_t - y_{t|t-1})^2] = \mathbb{E}[\eta_{t|t-1}^2]$
- $\beta_{t|T}$: expectation of β_t conditional on information up to T , i.e., the whole sample
- $P_{t|T}$: covariance matrix of β_t conditional on information up to T , i.e., the whole sample $\mathbb{E}[(\beta_t - \beta_{t|T})(\beta_t - \beta_{t|T})']$

4.2 Assumptions for now

- $\beta_{0|0}$ and $P_{0|0}$ are known (i.e., the initial state and its covariance)
- We will also, for now, assume that the parameter matrices, H , A , $\tilde{\mu}$, F , R and Q are known, and that z_t is known for all t

4.3 Prediction, updating, Smoothing

4.3.1 Prediction

At the beginning of time t we want to form an optimal predictor of y_t based on all the available information up to time $t - 1$, i.e., we want to calculate $y_{t|t-1}$. We start by predicting the state:

$$\beta_{t|t-1} = \tilde{\mu} + F\beta_{t-1|t-1}$$

with conditional covariance:

$$P_{t|t-1} = FP_{t-1|t-1}F' + Q$$

Accordingly, the condition forecast of y_t is:

$$y_{t|t-1} = H\beta_{t|t-1} + Az_t$$

4.3.2 Updating

Once y_t is realized at the end of time t we can calculate the forecast error, $\eta_{t|t-1}$ and its associated conditional covariance $f_{t|t-1}$:

$$\begin{aligned}\eta_{t|t-1} &= y_t - (H\beta_{t|t-1} + Az_t) \\ f_{t|t-1} &= HP_{t|t-1}H' + R\end{aligned}$$

The forecast errors ($\eta_{t|t-1}$) contain new information about the unobserved state β_t beyond that contained in $\beta_{t|t-1}$. Thus, we can use the new information contained in the forecast errors to update our belief about β_t , which is the goal of the Kalman Filter recursions.

The updating equations for $\beta_{t|t}$ and $P_{t|t}$ follow from employing the multivariate conditional normal distribution, and are:

$$\begin{aligned}\beta_{t|t} &= \beta_{t|t-1} + K_t\eta_{t|t-1} \\ P_{t|t} &= P_{t|t-1} - K_tHP_{t|t-1}\end{aligned}$$

where

$$K_t = P_{t|t-1}H'f_{t|t-1}^{-1} \tag{4}$$

is called the Kalman gain

4.3.3 Updating and intuition

- $\beta_{t|t}$ is formed as a kind of weighted average of $\beta_{t|t-1}$ and the new information contained in the forecast error. The weight put on the forecast error is determined by K_t , the Kalman gain
- Writing out (4) using the expression for $f_{t|t-1}$, and for simplicity, assuming that $\beta_{t|t}$ and H have dimensions (1×1) , we get:

$$K_t = \frac{1}{H} \frac{P_{t|t-1} H^2}{P_{t|t-1} H^2 + R}$$

an inverse function of R , the covariance of e_t , and a positive function of the uncertainty regarding $\beta_{t|t-1}$, i.e., $P_{t|t-1}$

4.3.4 Prediction and updating summary

1. Set $t = 1$
2. Prediction step:

$$\begin{aligned}\beta_{t|t-1} &= \tilde{\mu} + G\beta_{t-1|t-1} \\ P_{t|t-1} &= F P_{t-1|t-1} F' + Q\end{aligned}$$

3. Updating step

$$\begin{aligned}\eta_{t|t-1} &= y_t - H\beta_{t|t-1} - Az_t \\ f_{t|t-1} &= H P_{t|t-1} H' + R \\ \beta_{t|t} &= \beta_{t|t-1} + K_t \eta_{t|t-1} \\ P_{t|t} &= P_{t|t-1} - K_t H P_{t|t-1}\end{aligned}$$

where

$$K_t = P_{t|t-1} H' f_{t|t-1}^{-1}$$

4. set $t = t + 1$, return to 2

4.3.5 Smoothing

- Notice, the recursive filtering algorithm only uses information up to time t on every recursion
- Smoothing is an iteration that runs backward in time, starting from time $t = T$ and going back to time $t = 1$
- A smoothed estimate, denoted $\beta_{t|T}$, might provide us with a more accurate estimate of the unobserved state, simple because it uses more information than the basic filter, which gives us $\beta_{t|t}$
- Once the basic Kalman filter recursions have been run, we can simply iterate the two following equations backwards for $t = T, T - 1, \dots, 1$ to get $\beta_{t|T}$:

$$\begin{aligned}\beta_{t|T} &= \beta_{t|t} + P_{t|t} F' P_{t+1|t}^{-1} (\beta_{t+1|T} - F\beta_{t|t} - \tilde{\mu}) \\ P_{t|T} &= P_{t|t} + P_{t|t} F' P_{t+1|t}^{-1} (P_{t+1|T} - P_{t+1|t}) P_{t+1|t}^{-1'} F P_{t|t}'\end{aligned}$$

4.4 Missing values, forecasting and starting values

- If the observable vector y_t contains missing values for some or more of the n variables, the Kalman filter machinery can still be used
- When all observations are missing – the updating step will not be conducted
- If only some observations are missing, updating will be done, but only conditional on the information in the observed observations
- Thus, the Kalman filter can easily be used for inferring missing values in the observables, and for conditional forecasting
- To run the Kalman filter recursions for $t = 1, 2, \dots, T$, we need some starting values – we need $\beta_{0|0}$ and $P_{0|0}$
- In choosing $\beta_{0|0}$ and $P_{0|0}$ we will separate between two cases:
 - For stationary state variable processes, the unconditional mean and covariance matrix of β_t may be employed as $\beta_{0|0}$ and $P_{0|0}$, respectively. That is: $\beta_{0|0} = (I_k - F)^{-1}\tilde{\mu}$ and $\text{vec}(P_{0|0}) = (I_k - F \otimes F)^{-1}\text{vec}(Q)$
 - Second, for nonstationary state variable processes, the unconditional mean and covariance matrix of β_t do not exist. Accordingly, $\beta_{0|0}$ can be set at any arbitrary $(k \times 1)$ vector (i.e., wild guessing). To assign a very large uncertainty to this guess, we should assign very large values to the diagonal elements of $P_{0|0}$.

5 Filtering examples

Figure: Filtering, smoothing and forecasting using the Kalman Filter

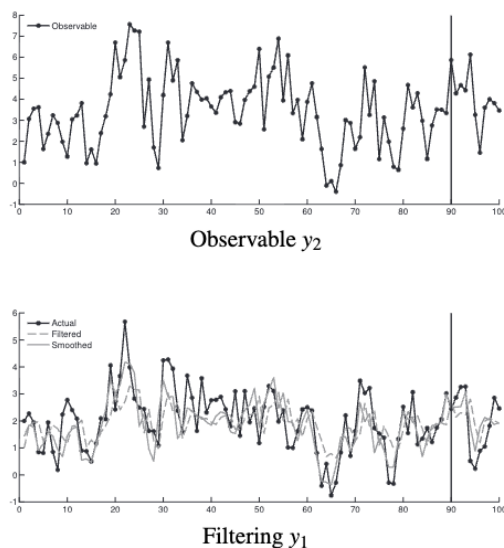
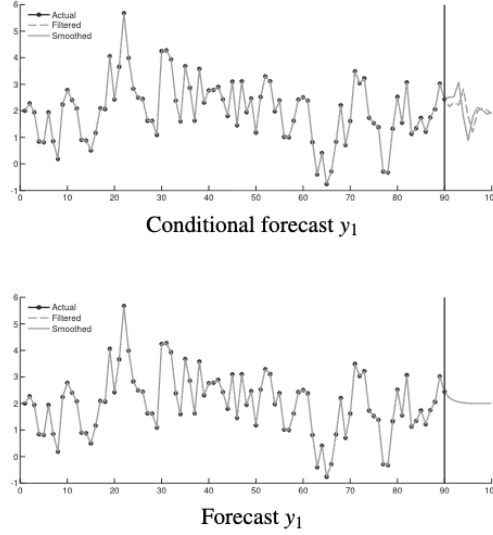


Figure: Filtering, smoothing and forecasting using the Kalman Filter



6 Kalman filter and MLE

- Up until now assumed that H , A , $\tilde{\mu}$, F , R and Q are known
- The Kalman filter can also be used to get the maximum likelihood estimate of these parameters
- To do so we utilize the prediction error decomposition of the log likelihood function:

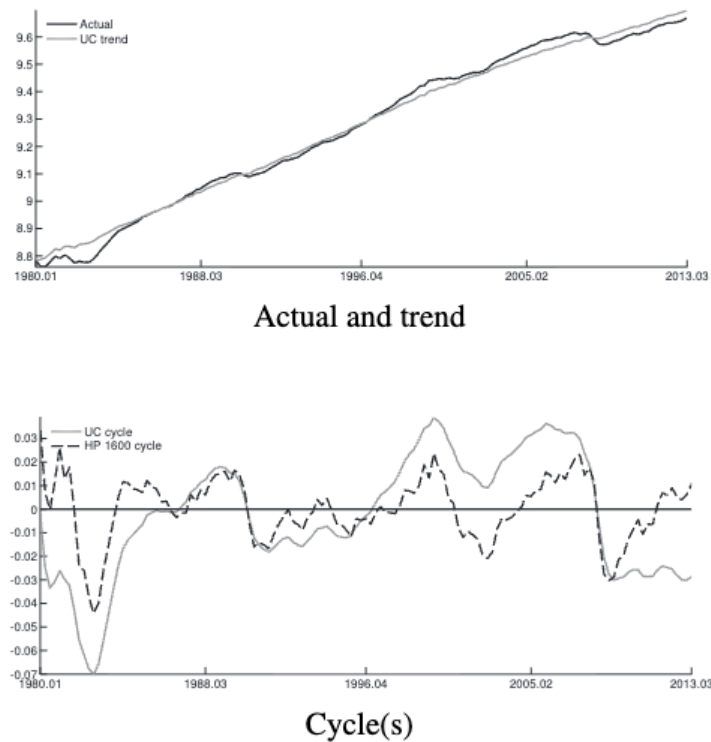
$$\ln L(\theta|\tilde{y}_T) = \sum_{t=1}^T \left[-\frac{1}{2} \ln((2\pi)^n |f_t|) - \frac{1}{2} \sum_{t=1}^T (\eta'_t f_t^{-1} \eta_t) \right] \quad (5)$$

- (5) is a function of η_t and f_t – derived within the KF iterations
- Thus, at each iteration of the filter, we can compute:

$$l(\theta) = l(\theta) - \frac{1}{2} \ln((2\pi)^n |f_{t-1}|) - \frac{1}{2} (\eta'_{t|t-1} f_{t|t-1}^{-1} \eta_{t|t-1})$$

7 The Kalman filter and MLE example

Figure: Estimation results: UC decomposition of US real GDP



8 Summary

1. The state-space representation consists of an observation equation and a transition equation. Nearly all linear models in economics can be put into a state-space representation
2. The Kalman filter is a recursive filter that can be used for estimating unobserved state variables, infer missing values among the observed variables, forecasting, and parameter estimation.
3. The Kalman filter consists of two steps: a prediction step followed by an updating step. Sometimes a third step is also added to the procedure, namely a smoothing step
4. When the parameters of the model are unknown, the Kalman filter together with the state-space representation can be used for MLE